Example 8.10

Consider $g(x,y)=\dfrac{2x^2y}{x^2+y^2}$ if $(x,y)\neq(0,0)$ and $g(0,0)=0$. Show that g is continuous on $\mathbb{R}^2$.

Solution

Observe that the function g is defined for all $(x,y)\in\mathbb{R}^2$. It is easy to check, as in the above examples, that g is continuous at all point $(x,y)\neq(0,0)$. Next we shall check the continuity of g at $(0,0)$. For that we see if g has a limit L at $(0,0)$ and if $L=g(0,0)=0$. So we consider

$$|g(x,y)-g(0,0)|=\left|\frac{2x^2y}{x^2+y^2}-0\right|=\frac{2|x^2y|}{|x^2+y^2|}=\frac{2|xy||x|}{x^2+y^2}\leq\frac{(x^2+y^2)|x|}{x^2+y^2}\leq|x| \qquad \ldots (9)$$

Note that in the final step above we have used $2|xy|\leq x^2+y^2$ (which follows by considering $0\leq(x-y)^2$) for all $x,y\in\mathbb{R}$. Note that $(x,y)\to(0,0)$ implies $|x|\to 0$. Then from (9) it follows that $\lim\limits_{(x,y)\to(0,0)}\dfrac{2x^2y}{x^2+y^2}=0=g(0,0)$; which proves that g is continuous at $(0,0)$. So g is continuous at every point of $\mathbb{R}^2$. ■

EXERCISE 8.3

1. Evaluate $\lim\limits_{(x,y)\to(1,2)}g(x,y)$, if the limit exists, where $g(x,y)=\dfrac{3x^2-xy}{x^2+y^2+3}$.

2. Evaluate $\lim\limits_{(x,y)\to(0,0)}\cos\left(\dfrac{x^3+y^2}{x+y+2}\right)$. If the limit exists.

3. Let $f(x,y)=\dfrac{y^2-xy}{\sqrt{x}-\sqrt{y}}$ for $(x,y)\neq(0,0)$. Show that $\lim\limits_{(x,y)\to(0,0)}f(x,y)=0$.

4. Evaluate $\lim\limits_{(x,y)\to(0,0)}\cos\left(\dfrac{e^x\sin y}{y}\right)$, if the limit exists.

5. Let $g(x,y)=\dfrac{x^2y}{x^4+y^2}$ for $(x,y)\neq(0,0)$ and $f(0,0)=0$.

   (i) Show that $\lim\limits_{(x,y)\to(0,0)}g(x,y)=0$ along every line $y=mx, m\in\mathbb{R}$.

   (ii) Show that $\lim\limits_{(x,y)\to(0,0)}g(x,y)=\dfrac{k}{1+k^2}$ along every parabola $y=kx^2, k\in\mathbb{R}\setminus\{0\}$.

6. Show that $f(x,y)=\dfrac{x^2-y^2}{y^2+1}$ is continuous at every $(x,y)\in\mathbb{R}^2$.

7. Let $g(x,y)=\dfrac{e^y\sin x}{x}$, for $x\neq 0$ and $g(0,0)=1$. Show that g is continuous at $(0,0)$.

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